

PROJECT SCOPE

Smart HR Management System

Feature Requirements and Expected Deliverables

Document Purpose

A concise implementation scope for the Backend, Frontend Web, Flutter Mobile, and AI workstreams. This document defines what each workstream must build and what is expected to be delivered, without prescribing any technology choices.

Product model	Employee Mobile App + HR/Manager Web Portal
Core principle	Employees perform daily actions on mobile; HR and managers control operations on web.
Document focus	Features, tasks, integration responsibilities, and deliverables
Version	September 2026

The system connects attendance, tasks, leave, goals, evaluations, performance, notifications, and AI-assisted development into one consistent employee lifecycle.

Project Summary

What the product is, who uses each surface, and how the workstreams connect.

The project is a smart HR management system designed around two main user experiences: a daily Employee Companion on mobile and an operational Control Center on the web for HR and managers. The Backend owns business rules, permissions, workflows, records, and integrations. The AI layer turns approved HR data into useful summaries, development guidance, policy answers, and explainable attention signals.

Workstream	Primary Users	Main Responsibility	Delivery Outcome
Flutter Mobile	Employees	Daily self-service: attendance, tasks, leave, goals, performance, AI, notifications, profile.	A complete employee-facing mobile flow that can be demonstrated end to end.
Frontend Web	HR, Managers, Admin	Manage people, teams, attendance, tasks, leave, evaluations, reports, policies, and AI insights.	A working control portal with role-based views and operational workflows.
Backend	All product surfaces	Own authentication, permissions, business logic, records, validation, file handling, notifications, and integrations.	Stable APIs and data workflows that support every approved mobile/web feature.
AI	Employees, Managers, HR	Generate grounded assistance and insights from allowed policies and employee data.	AI features that are explainable, permission-aware, useful, and safe for HR use.

Core Product Flows

1. Employee signs in, checks attendance status, and performs Check In using location validation.
2. Manager assigns a task; the employee updates progress, uploads a submission, and sends it for review.
3. Employee requests leave; the request is validated, reviewed, approved or rejected, and reflected in the employee record.
4. Goals, evaluations, attendance, and task outcomes contribute to the employee performance view.
5. AI uses only approved data to create development guidance, policy answers, performance explanations, and HR attention insights.

Product Boundary

The mobile app is employee-first. Full administration, workforce-wide reporting, policy management, and sensitive HR insights remain on the web portal.

Backend Features and Deliverables

The Backend is the source of truth for rules, data, permissions, workflows, and integrations.

Feature / Task	Required Behavior	Expected Delivery
Authentication & Sessions	Support sign in, sign out, password recovery, session renewal, account status checks, and authenticated user context.	Working authentication endpoints, session handling, error cases, and user identity response.
Roles & Permissions	Enforce Employee, Manager, HR, and Admin access rules on every protected operation.	Permission matrix implemented and verified across all feature endpoints.
Employee & Organization Data	Provide employee profile, department, manager, role, job information, status, and controlled profile updates.	Employee and organization APIs with validated create/read/update/deactivate operations.
Attendance	Receive Check In/Check Out requests, validate allowed location, prevent duplicates, calculate status and work duration, and return history.	Complete attendance workflow with validation, daily records, history, and clear failure reasons.
Company Locations	Store workplace locations and allowed attendance radius; expose only the data required for attendance.	Location management endpoints and attendance validation rules.
Leave Management	Create leave requests, validate dates and balances, route for review, approve/reject, update balance/history, and prevent conflicts.	End-to-end leave workflow including statuses, reviewer actions, history, and validation.
Task Management	Create and assign tasks, manage priority/deadline/status, update employee progress, and expose role-specific task lists.	Task lifecycle APIs supporting assignment through completion.
Task Submission & Review	Accept submission notes/files, create review state, support approve/reject/request changes, and preserve review history.	Submission workflow, review actions, status transitions, and audit history.
Goals	Store employee goals, target values, progress, target dates, ownership, and status changes.	Goal APIs with employee progress and manager/HR visibility.
Evaluations & Feedback	Store evaluation periods, category scores, manager feedback, overall results, and evaluation history.	Evaluation creation/retrieval flows with consistent scoring and access control.
Performance Aggregation	Prepare normalized performance data from attendance, tasks, goals, and evaluations for mobile/web views and AI requests.	Performance summary endpoint with period comparison and source metrics.
Notifications	Create in-app notification records and trigger push events for important workflow changes.	Notification center APIs, read/unread state, and event triggers.
Calendar Data	Expose a unified feed of approved leaves, holidays, task deadlines, and supported company events.	Calendar/events endpoint with date filtering.
File Handling	Validate file type/size, store secure file metadata, link files to requests or submissions, and enforce access permissions.	Upload/download flow with ownership checks and metadata records.
Company Policies	Store approved policy content and versions used by the HR Assistant.	Policy management and retrieval endpoints with active/version status.
AI Integration Gateway	Send only permitted, structured context to AI features and validate/store returned results when needed.	Secure AI request endpoints, permission checks, context assembly, and response logging.

Feature / Task	Required Behavior	Expected Delivery
Audit & Activity History	Record important administrative and workflow actions such as approvals, evaluations, status changes, and AI usage.	Queryable audit records for critical actions and troubleshooting.
Validation & Error Contract	Return consistent success/error structures and handle invalid states, missing permissions, duplicate operations, and bad input.	Documented API behavior and predictable error responses.

Backend Expected Handover

- All approved feature endpoints connected to persistent data and protected by role permissions.
- A clear API collection or API documentation covering request/response examples and error cases.
- Seed/demo data sufficient to demonstrate Employee, Manager, and HR scenarios.
- Validated workflow rules for attendance, leave, tasks, submissions, evaluations, notifications, and AI calls.
- Integration-ready environment for both web and mobile teams, with consistent endpoint behavior.

Frontend Web Features and Deliverables

The web portal is the operational workspace for HR, managers, and administrators.

Feature / Task	Required Behavior	Expected Delivery
Authentication & Role Routing	Provide secure sign in, session states, logout, protected routes, and role-aware navigation.	Usable login flow and correct access/navigation for HR, Manager, and Admin views.
HR Dashboard	Show workforce summary: employee counts, today attendance, lateness/absence, leave status, performance overview, pending reviews, and AI attention items.	Responsive dashboard with real API data, loading/empty/error states, and drill-down links.
Manager Dashboard	Show direct-team attendance, assigned tasks, overdue items, pending submissions, leave requests, evaluations, and team insights.	Team-focused dashboard restricted to the manager's authorized employees.
Employee Management	Search/filter employees, open profile, add/edit/deactivate employees, assign department/manager, and view employment details.	Complete employee management screens with forms, validation, and role controls.
Departments & Teams	View departments, members, managers, and basic organizational assignment.	Department/team management and assignment screens.
Attendance Management	View daily/monthly attendance, filter by employee/team/status, inspect records, and review attendance exceptions when allowed.	Attendance table, filters, detail view, and clear status visualization.
Leave Requests	List pending/history requests, inspect details/attachments, approve/reject, and show reviewer decision.	Operational leave review workflow with status filters and confirmation states.
Task Management	Create tasks, assign employees, set priority/deadline, monitor progress, filter tasks, and inspect task detail.	Task creation/management screens connected to live backend data.
Submission Review	Open employee submission, view note/attachments, and approve/reject/request changes with feedback.	Review queue and task-review detail workflow.
Goals Management	View employee/team goals, create or update allowed goals, and monitor progress against targets.	Goal lists, progress views, and editing where permitted.
Evaluations	Create evaluation, enter category scores and comments, view evaluation history, and mark evaluation complete.	Evaluation form, history, and employee/team evaluation views.
Performance Views	Show overall score, metric breakdown, trends, period comparison, and source indicators without hiding the underlying data.	Performance detail screen with understandable charts/tables and historical comparison.
AI Insights for HR/Managers	Display generated summaries, development gaps, evaluation draft assistance, and explainable employee attention signals only to authorized roles.	AI insight panels with evidence/factors, generated timestamp, and clear human-review language.
Policy Management	Add/update approved company policy content and control which policy version is active for employee assistance.	Policy list/editor and active-version control.
Notifications / Activity	Show relevant administrative activity and	Basic notification/activity center for web

Feature / Task	Required Behavior	Expected Delivery
	notification history where needed.	users.
Reports & Export	Provide practical filters and exportable views for attendance, leave, tasks, and performance.	Report pages with date/team/status filters and export action where approved.
UI States & Validation	Handle loading, empty results, errors, unauthorized access, confirmation dialogs, and form validation consistently.	Polished user experience across all core workflows, not only happy-path screens.

Frontend Web Expected Handover

- Complete HR and Manager workflows using real backend data rather than static mockups.
- Role-aware navigation and screen-level permission handling.
- Responsive layouts for common desktop/laptop sizes, with consistent components and states.
- Working forms, filters, tables, review actions, and performance/AI views.
- A demo-ready scenario covering employee setup, task assignment/review, leave approval, evaluation, and AI insight display.

Flutter Mobile Features and Deliverables

The mobile app is an Employee Companion: fast, focused, and useful every working day.

Feature / Task	Required Behavior	Expected Delivery
Splash & Session Check	Open the app, restore an active session when valid, and route the employee correctly.	Stable app start behavior with authenticated and unauthenticated states.
Login & Account Recovery	Support employee sign in, errors, password recovery entry point, and optional biometric re-entry after a valid session exists.	Complete login experience connected to backend authentication.
Home Dashboard	Show greeting, attendance state, worked time, task counts, next deadline, performance snapshot, and quick actions.	Employee home screen populated from live data with refresh/loading/error states.
Attendance Check In / Out	Request location only when needed, show workplace/distance status, submit attendance action, and display success/failure reason.	Working location-based Check In/Check Out flow against backend rules.
Attendance History	Show monthly attendance records with check-in/out time, status, and work duration.	Filterable monthly attendance history and day detail where available.
My Tasks	List tasks by status/priority/deadline, support filtering, and open task details.	Task list and detail screens with correct employee scope.
Task Progress	Allow employee to update progress/status within permitted transitions.	Progress update action reflected immediately in task details/history.
Task Submission	Allow submission note and approved attachments, then show review status and requested changes.	End-to-end task submission flow including upload state and review result.
Leave Request	Select leave type/dates, enter reason, add attachment if required, submit, and show validation errors.	Working request form connected to leave rules.
Leave History & Balance	Show request statuses, decision details, and available balance where supported.	Leave history and current balance view.
Goals	Show employee goals, progress, target, target date, and status.	Goal progress screen using live backend data.
My Performance	Show overall performance, metric breakdown, trend, and period comparison in an employee-safe form.	Performance screen with understandable metrics and trend presentation.
AI Career Coach	Request and display grounded strengths, development areas, and a practical development plan based on approved employee data.	Career Coach screen with generated plan, context-aware actions, and safe fallback states.
AI HR Assistant	Chat for company-policy questions and supported personal HR facts such as leave balance, with answers grounded in approved sources.	Chat experience with source/policy references where applicable and clear unsupported-answer behavior.
Notifications	Receive important events and show in-app notification history with read/unread state.	Notification inbox and deep links into relevant task/leave/evaluation screens where possible.
Calendar	Show approved leaves, holidays, task deadlines, and supported employee events in a simple calendar/list view.	Unified employee calendar connected to backend event feed.

Feature / Task	Required Behavior	Expected Delivery
Profile & Help	Show employee identity, job/department/manager, joined date, selected summaries, HR contact/help, policies, and logout.	Profile/help area with read-only HR-controlled fields and allowed self-service actions.
Permissions & Device States	Handle location permission denied, network loss, upload failure, notification permission state, and session expiry gracefully.	Reliable device/error-state handling suitable for demo and normal usage.

Flutter Expected Handover

- A complete employee journey from login to daily attendance, tasks, leave, performance, AI, notifications, and profile.
- Real backend integration for all MVP screens; no hard-coded final demo data.
- Location permission and attendance behavior tested for both inside and outside the allowed workplace radius.
- File upload, task submission, leave request, notification, and AI response states handled visibly.
- A release-ready demo build plus a documented list of any non-MVP screens or known limitations.

AI Features and Deliverables

AI should assist employees and HR with grounded, explainable outputs - not make automatic employment decisions.

AI Design Rule

Every AI result must be based on approved system data or approved company policies. The AI must not invent employee facts, must not expose restricted HR data to employees, and must not make final decisions about promotion, termination, compensation, or employee status.

Feature / Task	Required Behavior	Expected Delivery
AI Career Coach - Employee	Analyze allowed employee performance, goals, skills, task outcomes, and evaluation themes. Return strengths, development areas, and a practical plan with short-term actions.	Structured development plan with evidence-based reasons, prioritized actions, and a generated timestamp.
HR Policy Assistant - Employee	Answer questions using active company policies. For supported personal questions, combine policy rules with backend-provided facts such as leave balance.	Grounded answer, policy/source reference, unsupported-question fallback, and no cross-employee data exposure.
Performance Insight Generator	Convert structured performance metrics and trends into a short explanation: what improved, what declined, likely contributing work indicators, and what to review next.	Readable summary tied to actual metric values/periods; no invented causes stated as facts.
Skill-Gap & Development Recommendations	Use evaluations, goals, skill records, and task evidence to identify development gaps and recommend measurable improvement actions.	Prioritized skill gaps with recommended actions and suggested follow-up measures.
Evaluation Draft Assistant - Manager	Create a draft evaluation narrative from manager-entered scores, feedback notes, goals, and relevant work outcomes. The manager must edit/approve before saving.	Editable draft with strengths, improvement areas, and evidence prompts; never auto-submit an evaluation.
Team Insight Summary - Manager	Summarize team-level patterns such as overdue workload, completion trends, common skill gaps, and evaluation themes without exposing unnecessary personal data.	Short team summary with drill-down factors and clearly separated individual/team findings.
Employee Attention Signal - HR/Manager	Identify employees whose recent work indicators show meaningful decline or unusual change across attendance, task completion, goals, or evaluation trend. Use Low/Medium/High attention level rather than predicting resignation.	Explainable attention level, contributing indicators, comparison period, and recommended human follow-up action.
Insight History & Regeneration	Keep important generated insights with date/context version where the product needs comparison or auditability.	Stored insight metadata and ability to regenerate when source data changes.
AI Feedback Capture	Allow authorized users to mark an AI output as useful/not useful or flag it for review.	Simple feedback signal stored for quality review and demo evidence.
Permission-Aware Context	Build different context for Employee, Manager, and HR requests so each role sees only permitted fields and outputs.	Verified context rules and test cases proving restricted data is not included in employee-facing AI requests.
Safe Fallback Behavior	When data is missing, conflicting, or insufficient, return a clear limitation instead of confident speculation.	Consistent "insufficient data / unsupported request" response pattern.

AI Output Requirements

What “good” AI delivery should look like in this project.

1. Career Coach Output

- Overall development focus for the next review period.
- Top 2-3 strengths supported by recent evidence.
- Top 2-3 development areas supported by evaluation/task/goal signals.
- A short action plan with measurable actions, not generic motivational text.
- Suggested follow-up date or milestone for checking progress.

2. Policy Assistant Output

- Direct answer first, then a short explanation.
- Reference to the relevant approved policy or policy section whenever the answer comes from policy content.
- Personal HR facts only when supplied by the Backend for the signed-in employee.
- Clear refusal/fallback for questions outside the available policy scope or the user’s permissions.

3. Performance Insight Output

- State the period being analyzed and the actual metrics used.
- Separate observed facts from suggested interpretation.
- Explain trend direction rather than only returning a single score.
- Recommend a human action such as a one-to-one, goal adjustment, workload review, or development plan where appropriate.

4. Employee Attention Signal

- Use “attention level” or “early warning” language, not “this employee will leave.”
- Show contributing indicators such as repeated lateness trend, falling completion rate, missed goals, or evaluation decline.
- Compare against the employee’s own recent history where possible instead of making unfair workforce-wide assumptions.
- Never trigger an employment action automatically. The result is a prompt for HR/manager review only.

Recommended AI MVP

Career Coach + Policy Assistant + Performance Insight Generator + Evaluation Draft Assistant. The Employee Attention Signal can be included in the MVP only if its factors are explainable and the team has enough reliable historical/demo data; otherwise present it as a Phase 2 feature.

Cross-Team Integration and Acceptance

The project is complete only when the workstreams operate as one system.

Shared Integration Requirements

- Frontend Web and Flutter consume the same backend rules for authentication, permissions, statuses, dates, and validation.
- Backend remains the source of truth for business state. Client applications do not independently decide attendance validity, leave approval rules, or final task status.
- AI receives context through controlled backend requests rather than direct client access to sensitive records.
- Workflow status names are consistent across mobile, web, backend, and AI prompts/results.
- Every important feature has loading, success, empty, validation-error, authorization-error, and server-error handling.
- Demo accounts exist for at least one Employee, one Manager, and one HR user with realistic connected data.

Minimum End-to-End Demo Scenario

1. HR creates or activates an employee and assigns the employee to a manager.
2. Employee logs in on mobile and successfully checks in within the allowed workplace area.
3. Manager creates and assigns a task from the web portal.
4. Employee receives the task, updates progress, uploads a submission, and sends it for review.
5. Manager reviews the submission and completes an evaluation or feedback entry.
6. Employee submits a leave request; Manager or HR approves it; the employee sees the updated status.
7. Performance data updates and the Employee sees an employee-safe performance view.
8. AI generates a Career Coach plan for the Employee and an explainable Performance/Attention insight for an authorized Manager or HR user.

Acceptance Area	Expected Condition
Functional	All MVP workflows can be completed with real connected data and correct status transitions.
Permission	Employee cannot access other employees' records; Manager is limited to authorized team data; HR/Admin access follows defined rules.
Data Integrity	Repeated requests, invalid dates, invalid transitions, and missing records are handled without corrupting state.
UX	Core screens include loading, empty, validation, success, and error states.
AI Grounding	AI output clearly uses approved data/policies, handles missing context, and never fabricates personal facts.
AI Safety	Sensitive attention insights remain restricted; AI does not make final employment decisions.
Demo Readiness	A clean end-to-end scenario can be run without manually changing database records during the presentation.

MVP Delivery Checklist

Recommended minimum scope for a coherent first release and final project demonstration.

Backend MVP

- Authentication and role permissions
- Employee/department/manager data
- Location-based attendance and history
- Leave request and approval workflow
- Task assignment, progress, submission, and review
- Goals, evaluations, and performance aggregation
- Notifications and file handling
- Policy data and controlled AI gateway
- Audit records for critical actions

Frontend Web MVP

- HR/Manager dashboards
- Employee and department management
- Attendance monitoring
- Leave review
- Task assignment and submission review
- Goals and evaluations
- Performance detail
- AI insight panels
- Policy management
- Role-aware navigation and common UX states

Flutter MVP

- Login and session handling
- Home dashboard
- Attendance Check In/Out and history
- Task list/detail/progress/submission
- Leave request/history
- Goals
- My Performance
- AI Career Coach
- AI HR Assistant
- Notifications
- Profile/help

AI MVP

- Career Coach
- Policy Assistant

- Performance Insight Generator
- Evaluation Draft Assistant
- Permission-aware context
- Grounded/insufficient-data fallback
- Output logging or metadata sufficient for review

Optional / Phase 2

Advanced team analytics, Employee Attention Signal with historical trend calibration, richer reporting, calendar enhancements, additional AI quality feedback, and expanded automation can be added after the MVP workflows are stable.

Final Ownership Summary

A simple view of what each workstream is responsible for handing over.

Workstream	Final Expected Handover
Backend	A secure, documented, data-backed service layer implementing all approved business workflows and integrations.
Frontend Web	A complete HR/Manager operational portal connected to real APIs with role-aware workflows and insight views.
Flutter	A complete employee mobile experience connected to real APIs, including attendance, self-service, work tracking, and employee AI.
AI	Grounded, explainable, permission-aware assistance and insight features with safe fallback behavior and human review boundaries.

Definition of Done

The project should be considered ready for final delivery when the four workstreams are integrated, the core end-to-end scenario runs without manual intervention, role permissions are enforced, data remains consistent across mobile and web, and the AI outputs are grounded in approved data with clear human-review boundaries.

Final Product Positioning

A practical employee companion on mobile, an operational HR and manager control center on web, a reliable backend that owns the workflows, and an AI layer that improves clarity, development, and decision support without replacing human HR judgment.